Basics of Computer and Programming

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Computer:

- The word “computer” came from the word “compute”. According to oxford dictionary compute means “calculate the figure”.
- But now the computer is not just a calculator. Today computer perform various task.
- for ex : billing in super market, switching in telephone exchange, transaction in banks, operation in ATM, business in stock exchange and many more.
- Computer plays vital role in our life. Almost anything can be done by computer.

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- A computer may be defined as a programmable machine that can solve problems by accepting data, performing certain operation and presenting the results of those operations under the direction of detailed step by step instruction.
- Such a set of sequenced instructions, which cause a computer to perform particular operations is called PROGRAM.
- Basically a computer performs arithmetic and logical operation. Typically a computer perform following function:
  - It accepts information from the user through various input devices
  - It performs basic arithmetic and logic operations on the desired data.
  - It provides the desired output

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Impact of computer on society

- Business and Industry
- Health care
- Education
- Humanities
- Science and Technology
- Entertainment
- Government

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Characteristics of computer

All computers have certain common characteristics irrespective of their type and size.

- Speed
- Storage
- Accuracy
- Versatility
- Automation
- Diligence

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What is Computer Program

- A computer program is a set of instructions written in a programming language that a computer can execute to perform a specific task. These programs enable computers to process data, automate tasks, and solve problems.

Characteristics of a Computer Program:

- Logical sequence: Programs follow a well-defined sequence of steps.
- Written in a **programming language**: Programs are written using machine language, assembly language, or high-level languages.
- Requires translation: Before execution, programs must be converted into machine code that the computer understands.
- Stored in memory: Programs are stored in computer memory for execution.

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What is Programming Language:

- The **language** in which computer program is written is called Programming Language.
- **A programming language** is a set of rules and syntax used to write instructions (programs) that a computer can understand and execute.
- As computer hardware has progressed through different generations, the way we write instructions for it has also evolved. Programming languages have moved from machine language (very close to the hardware and written in binary) to assembly language (using simple symbolic codes), and finally to high-level languages (closer to human language, easier to learn and use).

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So, we divide programming language in three categories:

1. Machine language
2. Assembly language
3. High level language

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1. Machine Language (1st Generation):

- Machine language is the lowest-level programming language, consisting of binary code (0s and 1s) that a computer's CPU can directly execute.
- **Programming in machine language** originally meant physically setting switches and connecting wires to represent binary values (0s and 1s). This process was extremely tedious and made it impractical to create full-fledged programs.

Characteristics:

- Written in binary (0s and 1s).
- Directly executed by the CPU without translation.
- Difficult for humans to understand and write.
- Processor-dependent: Each CPU has its own machine language.

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Example of Machine Language (Binary Code):

(i) 6 + 5 : instruction is Add 6,5

machine code : 00000100 00000110 00000101

(ii) 6 – 5 : instruction is Sub 6,5

machine code : 00000010 00000110 00000101

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2. Assembly Language (2nd Generation) :

- Assembly language is a low-level language that uses **mnemonics (symbolic codes like ADD, SUB, MOV)** instead of binary instructions.
- Easier for humans to write, but still closely tied to hardware. Requires an assembler to translate assembly code into machine language.

Characteristics:

- Uses mnemonics (e.g., MOV, ADD, SUB) instead of binary.
- Easier to understand compared to machine language.
- Requires an assembler to convert into machine code.
- Processor-dependent: Different CPUs have different assembly languages.

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Example of Assembly Code:

```
MOV AL, 05h  ; Move 5 into register AL (AL=AL+5)

ADD AL, 02h  ; Add 2 to AL (AL=AL+2)
```

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3. High-Level Languages (3rd Generation and Beyond) :

- High-level languages are user-friendly programming languages that are independent of the computer architecture.
- They use English-like syntax and are easier to write, read, and maintain.
- HLL are generally Machine-independent and can run on different computer architectures with or without little modification.
- It is Easier to learn and debug than low-level languages. But it Requires compilation or interpretation to execute.

Characteristics:

- Designed to be closer to human language (e.g., English-like syntax in Qbasic, C, Python, Java).
- Focus more on problem-solving than hardware details.
- Requires a compiler or interpreter to convert the code into machine-readable form.

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Examples of High-Level Languages:

Procedural languages	:	Basic, Qbasic, C, Pascal, FORTRAN
Object-oriented languages	:	Java, C++, Python, dot net
Scripting languages	:	JavaScript, Python, PHP

Example of High-Level Code (Qbasic):

```
num1 = 5
num2 = 2
sum = num1 + num2
print "Sum:"; sum
```

This prints the sum of two numbers.

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Translators:

A **translator** is a program that converts code written in one form (human-friendly) into another form (machine-friendly).

There are three main types of translators:

1. **Assembler** :

   A translator that converts assembly language code into machine code. Because assembly is human-readable language for programmers but must be translated into binary (machine code) before a computer can run it.

   Example:

   Assembly: ADD AL, 5

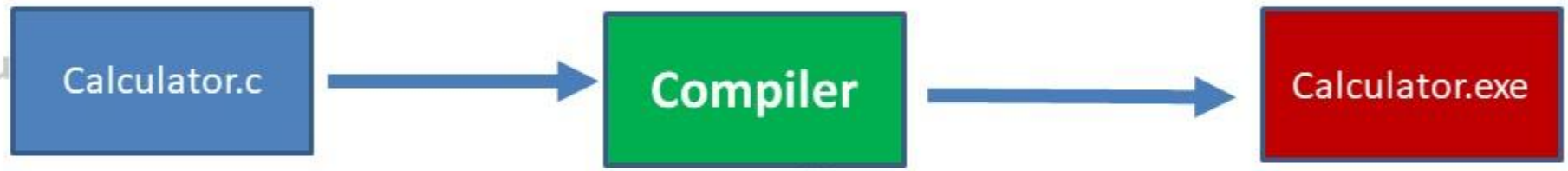
   Machine Code (Hex): 04 05

   Machine Code (Binary): 00000100 00000101

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2. Compiler

A translator that converts **high-level language** programs (like C, C++, Java etc) into **machine code** and save it in **executable file** in one go. These machine code is system dependent.



How it works:

- Reads the entire program.
- Translates it into a separate machine code file (executable).
- Then the executable can run multiple times without retranslation.

Advantages: Faster execution after compilation, errors are reported together.

Example: GCC (GNU Compiler for C/C++), javac (Java Compiler).

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3. Interpreter

- A translator that converts and executes high-level code line-by-line. But unlike compiler it does not create a executable file instead it send executable code directly to CPU for execution.
- So, every time to execute program interpreter need to interpret again line by line. Therefore interpreter language is slower than compiled language.



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How it works:

- Reads one instruction from the program.
- Translates it into machine code and send directly to CPU.
- Immediately runs it before moving to the next line.

Advantages: Easier debugging since errors appear immediately, no need for a separate executable.

Example: Qbasic interpreter, Java(JVM), Python Interpreter (CPython), JavaScript engines in browsers.

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